LEARNER GUIDE

How fast is the line changing?

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

The gradient of a line, m, tells you how steep it is and which way it leans. In this lesson the intercept is held fixed, so you can study m on its own. You change m and watch the line pivot about its crossing point — steeper, gentler, rising, falling, or flat. You connect the gradient to the constant step you met in a table (the rate of change), read and compare gradients, and model real rates like a plant growing or a tank draining. The question you are exploring is: what does m control when the intercept stays fixed?

What you are learning to notice

- That the gradient m is how much y changes each time x goes up by 1 — the rate of change.
- That the sign of m gives the direction: positive rises, negative falls, zero is flat.
- That the size of m gives the steepness: the bigger the size, the steeper the line.
- That changing m pivots the line about its intercept, which does not move.

Moving through it, one step at a time

In Gradient = the rate, change m and read the table beside the line — see the '1 across, m up' step. In the Gradient Isolator, the intercept is locked; drag m and watch the line pivot, rising, falling or flat. In Read and compare gradients, decide whether a line rises or falls and which of two lines is steeper. Then model a real rate — including a rate that goes down (a negative gradient). There is no rush; use the coordinate grid on the Scratchpad to draw the '1 across, m up' step for yourself.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading coordinates, and directed (positive and negative) numbers), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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